Around or Across? Radius, Diameter, Area, Perimeter Answer Key

Grades 6–7 Mathematics

Quick Answers

1. 37.70 cm
2. 78.54 in^2
3. 26 m
4. 35 ft^2
5. 43.98 m
6. (b) correct area = 201.06 cm^2, —
7. (b) correct amount = 36 ft^2, —
8. (a) trim length = 3.77 m, (b) glass area = 1.13 m^2

What is verified

6 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 6, 7. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 6–7 Mathematics
6.G.A – *problems 3, 4, 7*
7.G.B – *problems 1, 2, 5, 6, 8*
Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

1. *If they got 18.85: used C = pi times d but plugged in the radius, halving the answer*
2. *If they got 314.16: used the diameter 10 in place of the radius in A = pi r^2*
3. *If they got 36: computed the area 9 times 4 instead of the perimeter*
4. *If they got 24: found the perimeter of the rug instead of its area*
5. *If they got 87.96: treated the diameter 14 as a radius and used C = 2 pi r*
6. *If they got 804.25: used the diameter 16 in place of the radius in A = pi r^2*
7. *If they got 30: found the perimeter, but paint covers area*
8. *If they got 1.88: used C = pi times r, forgetting to double the radius*
8. *If they got 3.77: computed the circumference instead of the area*

1. Clock face, radius 6 cm: circumference.
Solution: Circumference uses the radius doubled: $C = 2\pi r = 2\pi \cdot 6 = 37.699\ldots$, so $\boxed{37.70 \text{ cm}}$

2. Pizza, diameter 10 in: area.
Solution: The area formula needs the *radius*: $r = 10 \div 2 = 5$. Then $A = \pi r^2 = \pi \cdot 5^2 = 78.539\ldots$, so $\boxed{78.54 \text{ in}^2}$

3. Garden 9 m by 4 m: fencing.
Solution: Fence runs *around* the garden, so we need perimeter, not area: $P = 2(9+4) = 2 \cdot 13$, giving $\boxed{26 \text{ m}}$

4. Rug 7 ft by 5 ft: floor covered.
Solution: "How much floor does it cover" asks for *area*: $A = 7 \cdot 5$, giving $\boxed{35 \text{ ft}^2}$
(as coordinates: the rug's corners $(0,0)$, $(7,0)$, $(7,5)$, $(0,5)$ enclose 35 square feet).

5. Pond, diameter 14 m: distance around.
Solution: With the *diameter* given, use $C = \pi d$ directly: $C = \pi \cdot 14 = 43.982\ldots$, so

$$\boxed{43.98 \text{ m}}$$

A common error is treating 14 as the radius and doubling again to get 87.96.

6. Mara's circle, diameter 16 cm.
Solution (a): *(instructor-judged)* Mara substituted the *diameter* 16 into $A = \pi r^2$. The radius is half the diameter: $r = 16 \div 2 = 8$ cm. Her 804.25 is four times too large.
Solution (b): $A = \pi \cdot 8^2 = \pi \cdot 64 = 201.061\ldots$, so $\boxed{201.06 \text{ cm}^2}$

7. Jon's paint for a 12 ft by 3 ft stripe.
Solution (a): *(instructor-judged)* $2(12+3) = 30$ is the *perimeter* — the distance around the stripe. Paint covers surface, so Jon needed the *area*. His 30 square feet is a perimeter wearing an area's unit.
Solution (b): $A = 12 \cdot 3$, giving $\boxed{36 \text{ ft}^2}$

8. Challenge: table top, radius 0.6 m — trim (around) and glass (covering).
Solution (a): Trim is circumference: $C = 2\pi r = 2\pi \cdot 0.6 = 3.769\ldots$, so $\boxed{3.77 \text{ m}}$
Solution (b): Glass is area: $A = \pi r^2 = \pi \cdot 0.6^2 = \pi \cdot 0.36 = 1.130\ldots$, so $\boxed{1.13 \text{ m}^2}$
Note the two answers use different units — meters for a length, square meters for a surface.